KGiSL MicroCollege
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TECHNICAL TEAM

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KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
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ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

CERTIFI

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COURSE NAME : B.Sc.CS
DEPARTMENT : B.Sc.CS-I-C
SUBJECT : JAVA PROGRAMMING

INTRODUCTION

1.Vehicle parking management is an essential system in modern urban areas.

2.It helps in organizing parking spaces efficiently.

3.Automated fee collection reduces manual errors and saves time.

4.Report generation helps in tracking daily, weekly,and monthly parking activities.

PROBLEM STATEMENT:

1.Manual parking systems are time-consuming and error-prone.
2.Difficulty in maintaining records of parked vehicles.
3.Revenue tracking is inefficient without automation.
4.Need for a system to manage parking, collect fees, and generate reports automatically.

OBJECTIVES OF THE PROJECT

1.Develop an automated vehicle parking system.
2.Implement parking fee calculation based on time duration.
3.Maintain records of vehicle entry and exit.
4.Generate reports for parking usage and revenue.

SCOPE OF THE SYSTEM

The system can be used in:

1.Shopping malls
2.Office buildings
3.Hospitals
4.Educational institutions
5.Public parking areas

SYSTEM MODULES

Module	Description
Vehicle entry	Record vehicle number and entry time
Vehicle exit	Calculate parking duration and fee
Fee collection	Collect and display parking charges
Report generation	Generate daily/monthly reports
Admin module	Ma

SYSTEM ARCHITECTURE /WORKFLOW

Start

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Vehicle Entry

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Store Vehicle Details

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Vehicle Exit

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Calculate Parking Fee

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Generate Report

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Logout

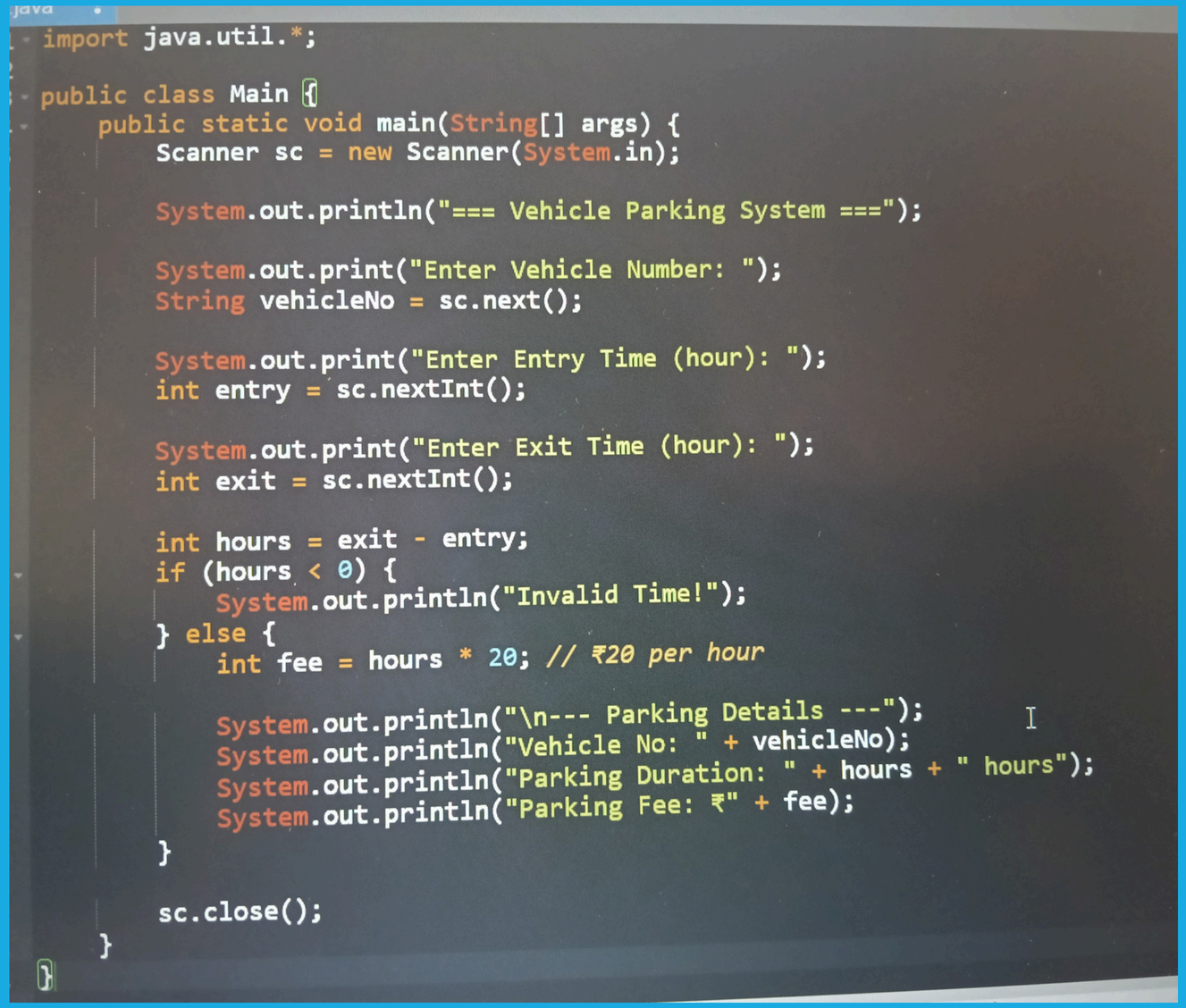
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End

JAVA CONCEPTS USED

1.Classes and Objects – Used to represent vehicles and parking system
2.Constructor – Initialize system components
3.Inheritance – Extend classes for GUI features
4.Event Handling – Handle button clicks for entry/exit
5.Exception Handling – Manage invalid inputs
6.String Handling – Process vehicle numbers
7.Time Calculation Logic – Calculate parking duration and fee
8.Java Swing (GUI) – JFrame, JButton, JTextField, JOptionPane
9.File Handling (Optional) – Store parking records

JAVA PROGRAM



```java
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("=== Vehicle Parking System ===");

        System.out.print("Enter Vehicle Number: ");
        String vehicleNo = sc.next();

        System.out.print("Enter Entry Time (hour): ");
        int entry = sc.nextInt();

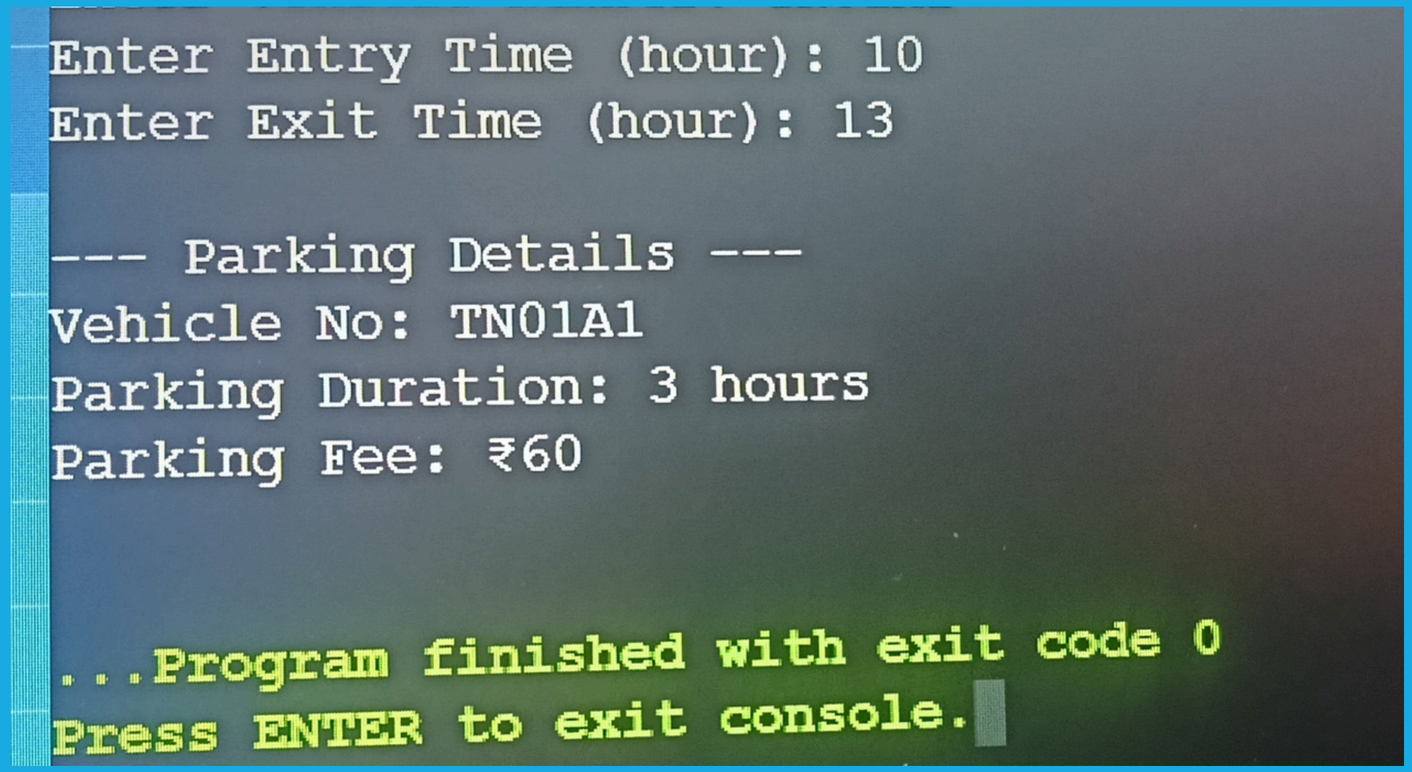
        System.out.print("Enter Exit Time (hour): ");
        int exit = sc.nextInt();

        int hours = exit - entry;
        if (hours < 0) {
            System.out.println("Invalid Time!");
        } else {
            int fee = hours * 20; // ₹20 per hour

            System.out.println("\n--- Parking Details ---");
            System.out.println("Vehicle No: " + vehicleNo);
            System.out.println("Parking Duration: " + hours + " hours");
            System.out.println("Parking Fee: ₹" + fee);
        }

        sc.close();
    }
}
```

OUTPUT SCREEN



DEMONSTRATION STEPS

1. Run the application
2. Enter vehicle details at entry
3. Record entry time
4. On exit, calculate parking fee
5. Display fee and collect payment
6. Generate report

CONCLUSION

1.The project successfully implements a vehicle parking system
2.Automates fee collection and record maintenance
3.Reduces manual work and improves accuracy
4.Demonstrates core Java programming concepts

FUTURE ENHANCEMENTS

1. Database integration (MySQL / Oracle)
2. Online booking of parking slots
3. QR code-based vehicle entry
4. Mobile application support
5. Real-time parking availability tracking